Equinumerosity is an equivalence relation

The Open Logic Project contributors

openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment equinumerosityisequi (source lines 1011-1013, the \ollabel on line 1011); the proof environment begins two lines below it and is its own proof, closing at source line 1032.

Editorial note. Neither the statement nor the proof contains an \olref or a \cite, so no source pointer and no bibliography entry are needed; the notation for equinumerosity and for the identity map used in the proof is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's size-of-sets unit, counted once: equinumerosity is an equivalence relation.

Allowed standard prerequisites (editorial).

- Equinumerosity. A is equinumerous with B when there is a bijection f from A to B.. This is the definition unfolded at each of the three steps (source lines 1019-1031).
- Closure properties of bijections. The identity map of a set is a bijection; the inverse of a bijection is a bijection; the composite of two composable bijections is a bijection.. The three clauses of the proof use exactly one of these three facts each (source lines 1019-1020, 1024-1026, 1029-1031).

Proposition 0.1. *Equinumerosity is an equivalence relation.*

Proof. We must show that equinumerosity is reflexive, symmetric, and transitive. Let A, B, and C be sets.

Reflexivity. The identity map $\mathrm{id}A\colon A \to A$, where $\mathrm{id}A(x) = x$ for all $x \in A$, is !!abijection. So $A = A$.

Symmetry. Suppose $A = B$, i.e., there is !!abijection $f\colon A \to B$. Since f is !!bijective, its inverse f^{-1} exists and is also !!bijective. Hence, $f^{-1}\colon B \to A$ is !!abijection, so $B = A$.

Transitivity. Suppose that $A = B$ and $B = C$, i.e., there are !!bijections $f\colon A \to B$ and $g\colon B \to C$. Then the composition $\circ fg\colon A \to C$ is !!bijective, so that $A = C$. □